

A bottom navigation indicator that gets hand-drawn.

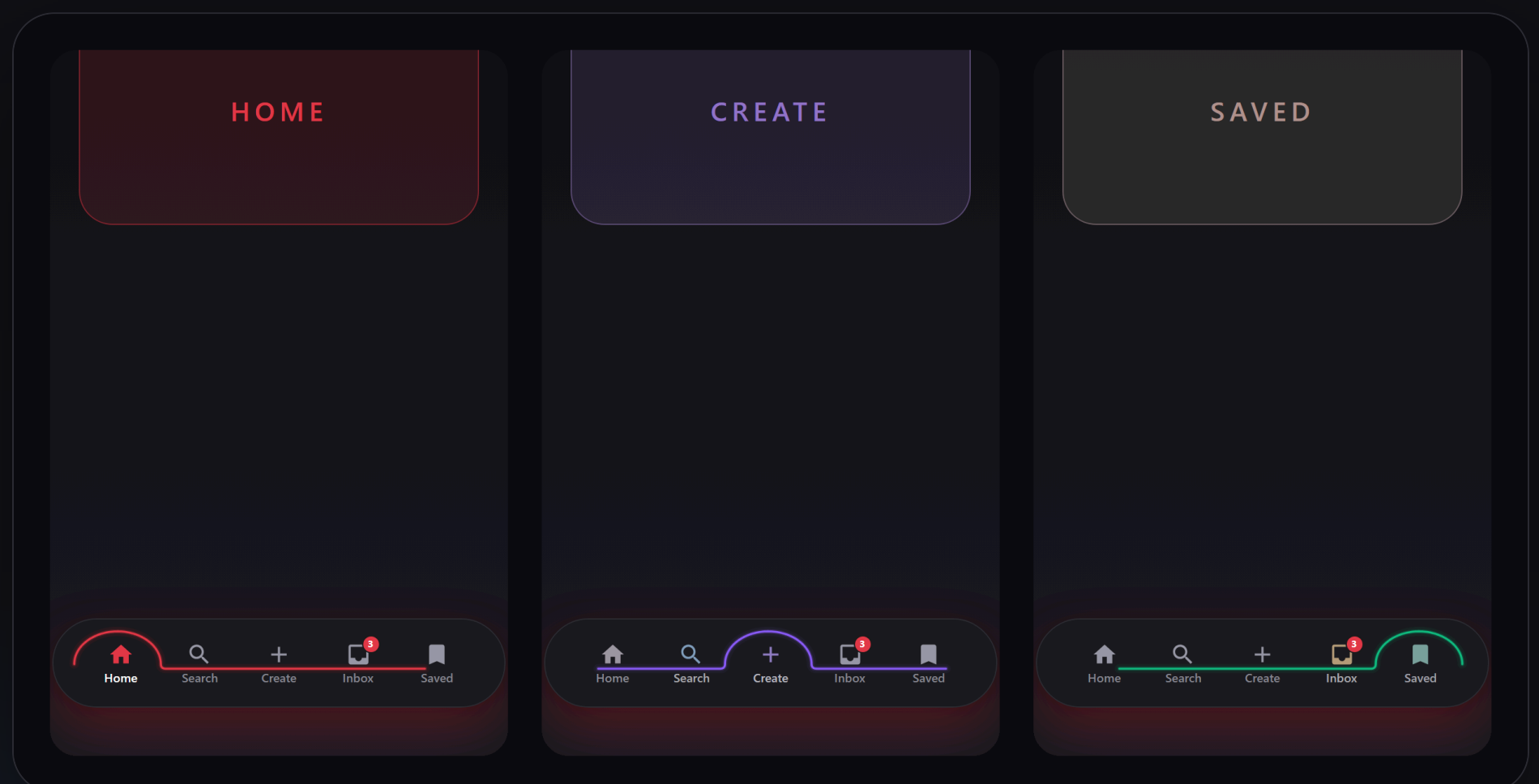
Organic Bézier curves, a path-reveal animation and one accent per destination — all in pure Compose Canvas.

Compose Canvas

PathMeasure

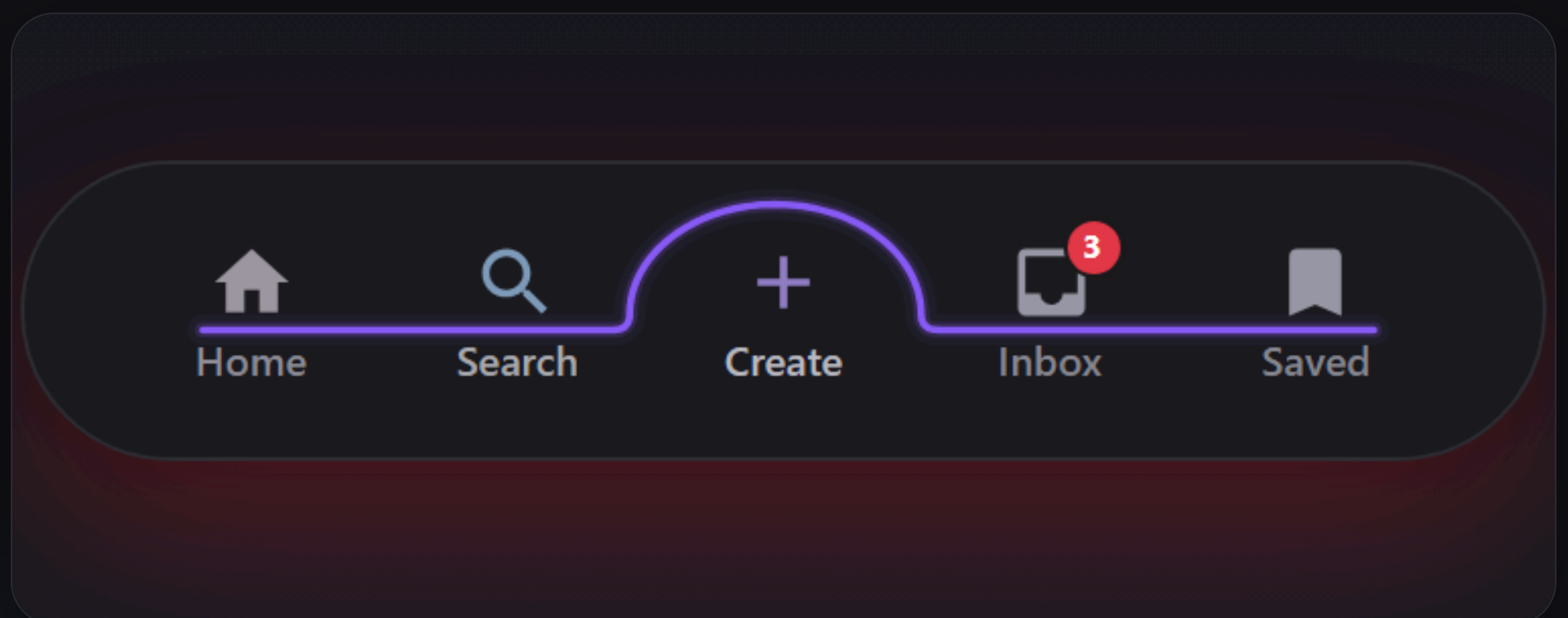
Navigation 3

Kotlin 2.2



Most bottom bars slide. This one is drawn.

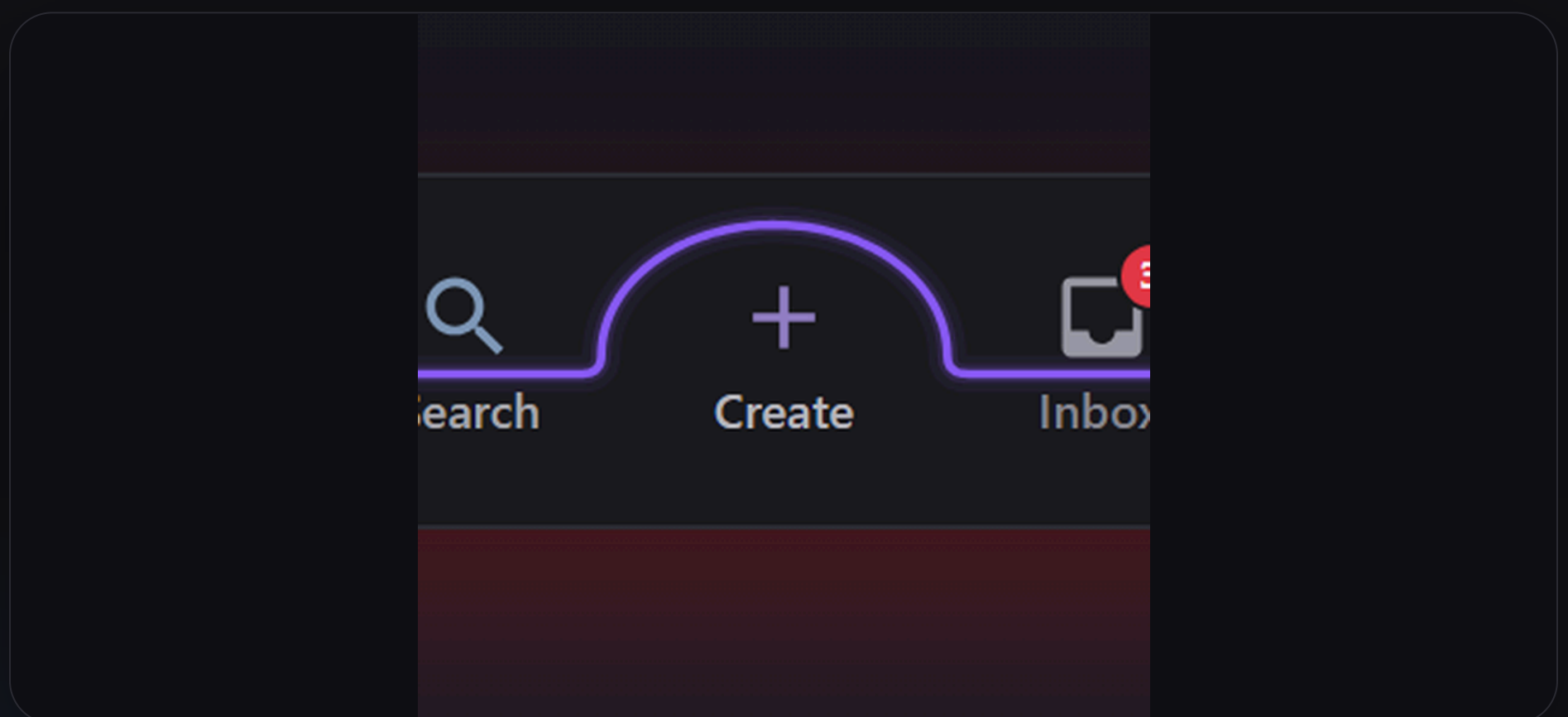
Instead of fading an icon or sliding a highlight, the selection is a single continuous stroke: an elliptical crown that wraps the active icon and trails off horizontally into the bar.



One Path · no corners · no flat top · no straight vertical edges

Anatomy of the indicator.

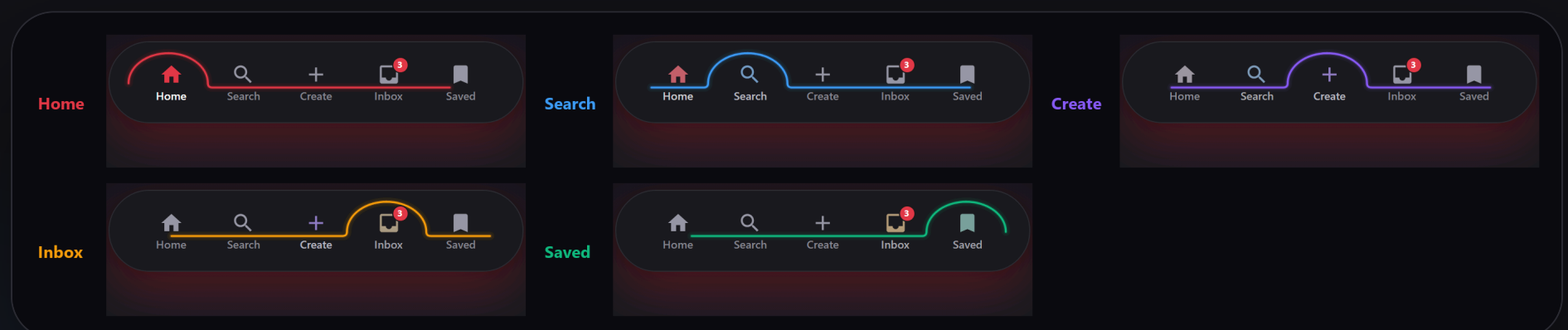
- ◆ **Elliptical crown** — two cubic Béziers with the quarter-arc constant $K = 4/3(\sqrt{2}-1) \approx 0.5523$.
- ◆ **Concave fillets** blend the ellipse sides into flat horizontal tails, so the curve never kinks.
- ◆ The ellipse is centered on the **measured icon center** — never the item or label center.
- ◆ Ellipse width = **0.62** × the selected cell, height from the measured content rect with **0.22** vertical padding.



Path-reveal, not fade.

- ◆ Geometry is fixed at the selected item; only the **drawn length** animates.
- ◆ **PathMeasure.getSegment()** walks the real path geometry.
- ◆ 500 ms
FastOutSlowInEasing; a fresh **Animatable** per selection cancels any in-flight reveal.
- ◆ The initial selection renders instantly — no flash on first frame.

```
// new Animatable per selection
val reveal = remember(selectedItemId) {
    Animatable(0f)
}
LaunchedEffect(selectedItemId) {
    reveal.animateTo(
        1f, tween(500, FastOutSlowInEasing)
    )
}
```



One accent, everywhere.

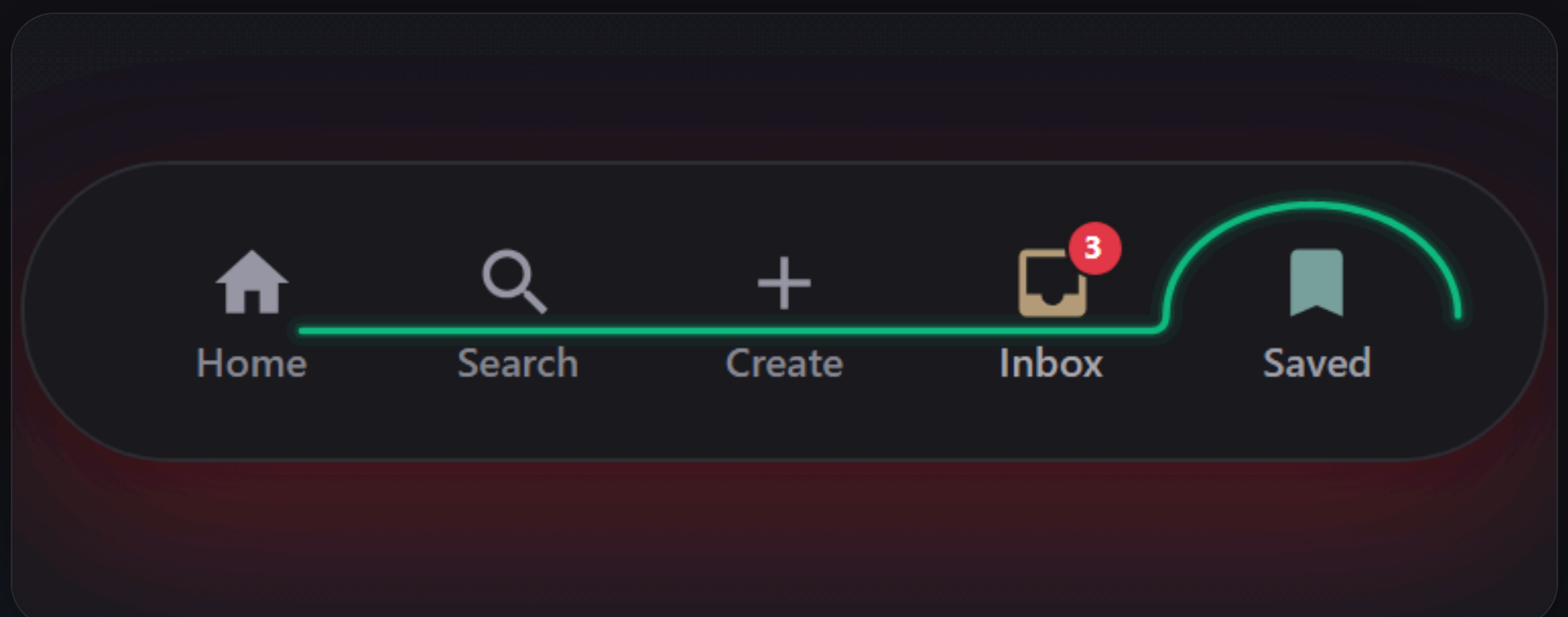
Each destination owns a color. That single value drives **both** the indicator stroke and the selected icon tint, so they can never drift apart. Unselected items stay neutral.

- Home · #E53946
- Search · #3D9DF6
- Create · #8B5CF6
- Inbox · #F59E0B (+ badge)
- Saved · #10B981

```
data class NavItem(  
    val id: String,  
    val icon: ImageVector,  
    val label: String,  
    val color: Color,           // stroke + icon tint  
    val badgeCount: Int? = null,  
)
```

Tail budget & edge trimming.

- ◆ The horizontal line budget **transfers from right to left** as the selection moves right, so their sum stays constant.
- ◆ The **first** item keeps only the right tail, the **last** only the left — the ellipse itself never moves.
- ◆ A leading **8%** of the curve is always trimmed; the trailing **8%** is trimmed only on the last item.
- ◆ Three stacked stroke passes (alpha 0.07 / 0.16 / 1.0) create the soft neon glow.



Wired to Navigation 3.

- ◆ **NavDisplay + NavBackStack + entryProvider** drive the screens.
- ◆ Tapping a tab clears the stack and pushes one route — single-level tabs.
- ◆ A custom **listSaver** persists the back stack across process death **without kotlin.serialization**.
- ◆ The bar owns no navigation state — the selected id is passed in.

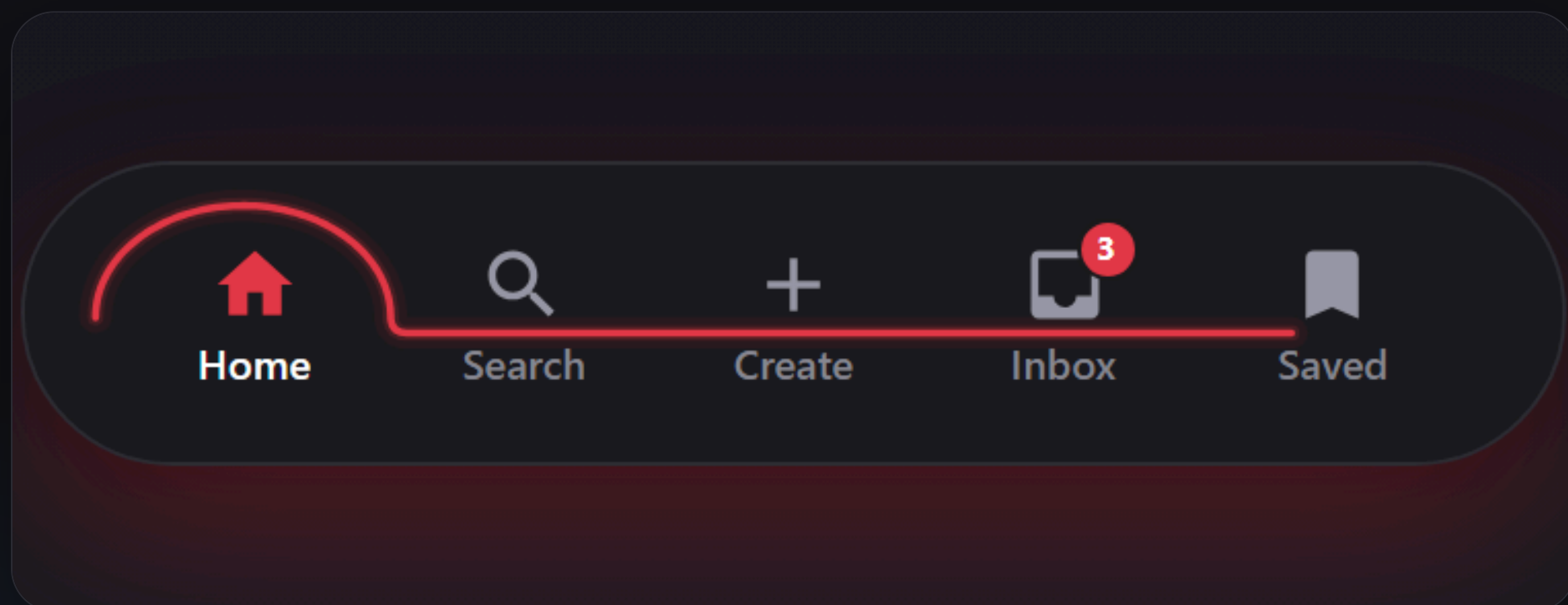
```
sealed interface AppNavKey : NavKey
data object HomeRoute : AppNavKey
data object SearchRoute : AppNavKey
data object CreateRoute : AppNavKey
data object InboxRoute : AppNavKey
data object SavedRoute : AppNavKey
```

NavDisplay

rememberSaveable

listSaver

edge-to-edge



TECH STACK

Built with

Kotlin 2.2.10

Compose BOM 2026.02.01

Navigation 3 · 1.1.6

Material 3

AGP 9.4

minSdk 26 · targetSdk 37

Zero icon libraries

100% Kotlin

Icons are compiled from raw SVG path strings at runtime, so the whole thing ships with no extra icon dependency.

Full source is on GitHub →

github.com/arad-sheybak/fancy_bottom_navigation

Star it if you like it, and tell me in the comments:

what should I build next — a curved app bar or a morphing FAB?